DAY 18 | PRE-READ

Saving Versions and Packing Boxes

A light read before Day 18. About 15 minutes, no code required.

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Why read this?

Tomorrow you learn three tools every software team uses daily: Git to track changes, pipelines to check them automatically, and Docker to package an app so it runs the same everywhere.

1. Git: save points for your work

Video games let you save progress so you can go back if something goes wrong. **Git** gives your code save points, called **commits**. Each one records what changed, who changed it and why. You can always return to an earlier version.

2. Branches: parallel drafts

When several people edit one document at the same time, it gets messy. Instead, each person works on their own copy, then the changes are carefully combined. In Git these copies are **branches**, and the request to combine them is a **pull request**, where teammates review the work first.

> **Key idea:** the main branch should always work. New work happens on branches and is reviewed before joining main.

3. Automatic checks on every change

Yesterday you met CI/CD. Tomorrow you write the pipeline file that tells Azure DevOps exactly which checks to run, in which order, every time someone pushes code.

4. Docker: packing the whole kitchen

Imagine shipping a recipe to a friend abroad. Their oven, ingredients and measuring cups are different, so the dish turns out differently. Now imagine shipping a sealed box containing the dish, the ingredients and the oven. That is a **container**. It runs the same on your laptop, a test server or the cloud.

Docker word	Plain meaning
Dockerfile	The recipe: step-by-step instructions to build the box
Image	The sealed box, built from the recipe
Container	The box opened and running
Registry	A warehouse that stores boxes so others can use them

5. Before class

- VS Code users: install Git and Docker Desktop if your laptop allows it. Colab users can do the Git parts in the notebook.
- If your organisation provides Azure DevOps access, sign in once to check it works.

6. Think about these before class

- Have you ever lost work because a file was overwritten? How would save points have helped?
- What would make a code review feel helpful rather than critical?
- Why might software that works on your laptop fail on someone else's?

Junbi (準備): preparation